

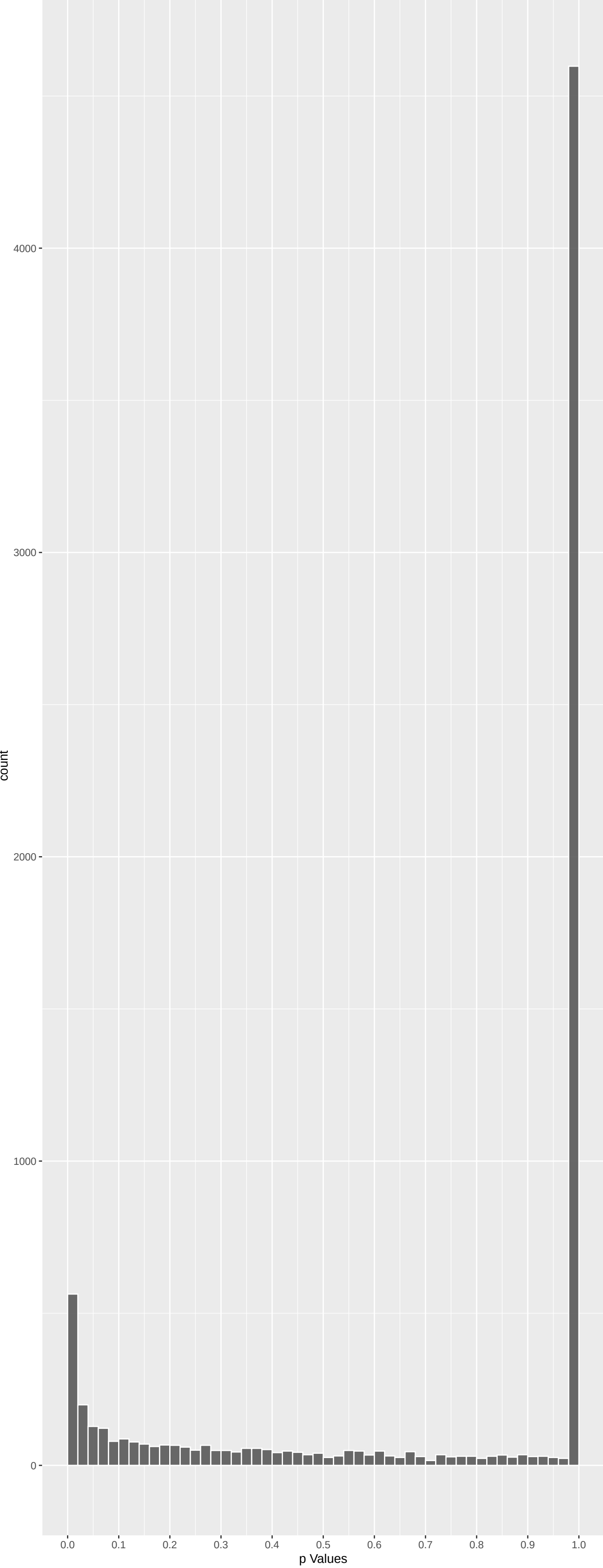
Top genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
MUC20	-6.156108	4.473250e-09	3.378e-05	2.552e-01
EP400	6.184803	3.730794e-09	3.378e-05	2.552e-01
LRPPRC	-5.863975	2.711497e-08	1.089e-04	2.741e-01
TGFBRAP1	5.816559	3.604273e-08	1.089e-04	2.741e-01
NEDD4	-5.825862	3.409113e-08	1.089e-04	2.741e-01
GOLGA4	-5.774340	4.635331e-08	1.167e-04	2.741e-01
PKD1L3	-5.733987	5.885818e-08	1.270e-04	2.741e-01
GALNT5	-5.619752	1.147389e-07	2.166e-04	2.867e-01
NEK5	-5.566300	1.561232e-07	2.620e-04	2.867e-01
LRIT1	5.541275	1.801659e-07	2.721e-04	2.867e-01
USP11	5.427154	3.435589e-07	3.227e-04	2.867e-01
PTCD3	-5.417216	3.632054e-07	3.227e-04	2.867e-01
LRRC53	-5.466043	2.761150e-07	3.227e-04	2.867e-01
TD RD7	5.417627	3.623723e-07	3.227e-04	2.867e-01
TRPM6	-5.422723	3.521878e-07	3.227e-04	2.867e-01
STOX1	-5.430804	3.366035e-07	3.227e-04	2.867e-01
MOV10L1	5.494474	2.351262e-07	3.227e-04	2.867e-01
KIAA0408	-5.388725	4.257557e-07	3.573e-04	2.988e-01
MBOAT4	-5.314738	6.408670e-07	4.852e-04	3.672e-01
KIAA2012	-5.313925	6.437318e-07	4.852e-04	3.672e-01
TNNI3K	-5.292734	7.230080e-07	5.200e-04	3.741e-01
RNF169	-5.273381	8.035991e-07	5.517e-04	3.788e-01
COL12A1	5.255170	8.873259e-07	5.827e-04	3.827e-01
OTOG	-5.227677	1.029919e-06	6.482e-04	3.907e-01
ZSCAN20	-5.205366	1.161685e-06	6.985e-04	3.907e-01

Top genes by Q-Value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
MUC20	-6.156108	4.473250e-09	3.378e-05	2.552e-01
EP400	6.184803	3.730794e-09	3.378e-05	2.552e-01
LRPPRC	-5.863975	2.711497e-08	1.089e-04	2.741e-01
GOLGA4	-5.774340	4.635331e-08	1.167e-04	2.741e-01
TGFBRAP1	5.816559	3.604273e-08	1.089e-04	2.741e-01
NEDD4	-5.825862	3.409113e-08	1.089e-04	2.741e-01
PKD1L3	-5.733987	5.885818e-08	1.270e-04	2.741e-01
USP11	5.427154	3.435589e-07	3.227e-04	2.867e-01
PTCD3	-5.417216	3.632054e-07	3.227e-04	2.867e-01
GALNT5	-5.619752	1.147389e-07	2.166e-04	2.867e-01
LRRC53	-5.466043	2.761150e-07	3.227e-04	2.867e-01
TD RD7	5.417627	3.623723e-07	3.227e-04	2.867e-01
TRPM6	-5.422723	3.521878e-07	3.227e-04	2.867e-01
LRIT1	5.541275	1.801659e-07	2.721e-04	2.867e-01
STOX1	-5.430804	3.366035e-07	3.227e-04	2.867e-01
MOV10L1	5.494474	2.351262e-07	3.227e-04	2.867e-01
NEK5	-5.566300	1.561232e-07	2.620e-04	2.867e-01
KIAA0408	-5.388725	4.257557e-07	3.573e-04	2.988e-01
MBOAT4	-5.314738	6.408670e-07	4.852e-04	3.672e-01
KIAA2012	-5.313925	6.437318e-07	4.852e-04	3.672e-01
TNNI3K	-5.292734	7.230080e-07	5.200e-04	3.741e-01
RNF169	-5.273381	8.035991e-07	5.517e-04	3.788e-01
COL12A1	5.255170	8.873259e-07	5.827e-04	3.827e-01
PARP12	5.190157	1.260701e-06	7.053e-04	3.907e-01
ZSCAN20	-5.205366	1.161685e-06	6.985e-04	3.907e-01

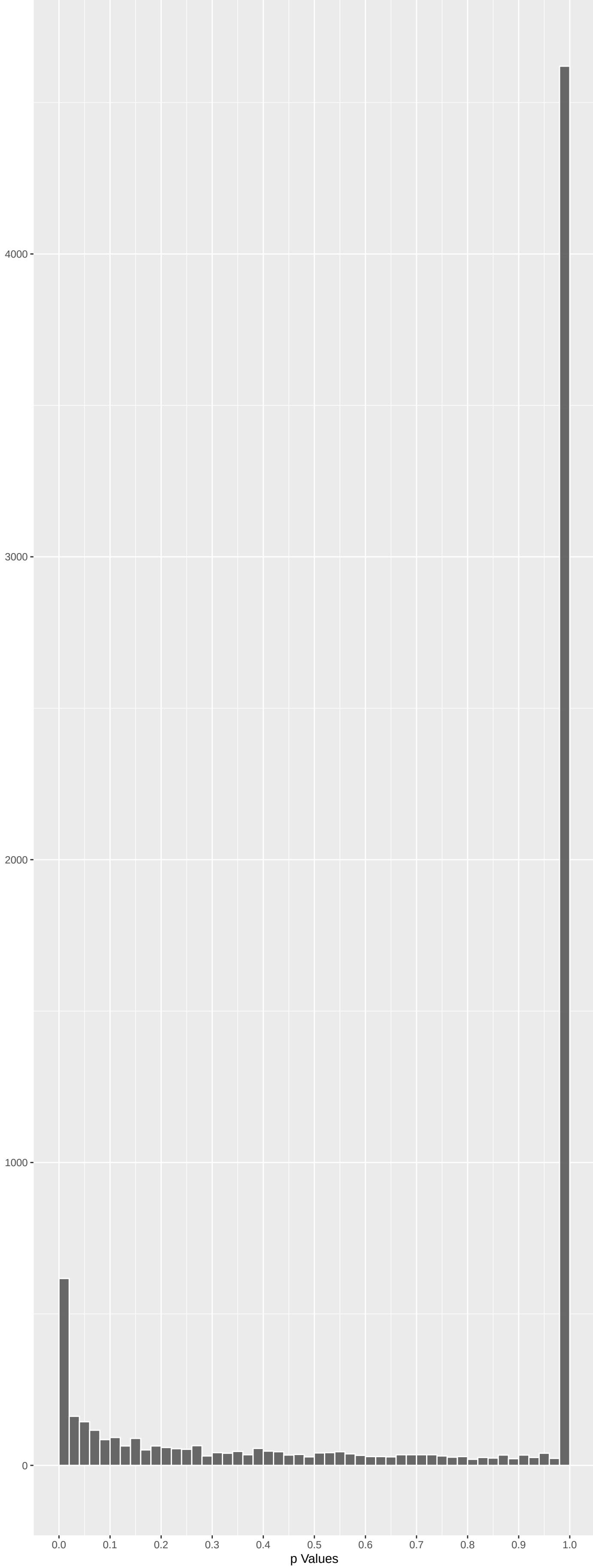
Positive Rho Non-permulated



Top Positive genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
EP400	6.184803	3.730794e-09	3.378e-05	2.552e-01
TGFBRAP1	5.816559	3.604273e-08	1.089e-04	2.741e-01
LRIT1	5.541275	1.801659e-07	2.721e-04	2.867e-01
USP11	5.427154	3.435589e-07	3.227e-04	2.867e-01
TD RD7	5.417627	3.623723e-07	3.227e-04	2.867e-01
MOV10L1	5.494474	2.351262e-07	3.227e-04	2.867e-01
COL12A1	5.255170	8.873259e-07	5.827e-04	3.827e-01
PARP12	5.190157	1.260701e-06	7.053e-04	3.907e-01
FRMPD4	5.178433	1.342542e-06	7.243e-04	3.907e-01
NSD2	5.137827	1.667599e-06	8.396e-04	4.228e-01

Negative Rho Non-permulated



Top Negative genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
MUC20	-6.156108	4.473250e-09	3.378e-05	2.552e-01
LRPPRC	-5.863975	2.711497e-08	1.089e-04	2.741e-01
NEDD4	-5.825862	3.409113e-08	1.089e-04	2.741e-01
GOLGA4	-5.774340	4.635331e-08	1.167e-04	2.741e-01
PKD1L3	-5.733987	5.885818e-08	1.270e-04	2.741e-01
GALNT5	-5.619752	1.147389e-07	2.166e-04	2.867e-01
NEK5	-5.566300	1.561232e-07	2.620e-04	2.867e-01
PTCD3	-5.417216	3.632054e-07	3.227e-04	2.867e-01
LRRC53	-5.466043	2.761150e-07	3.227e-04	2.867e-01
TRPM6	-5.422723	3.521878e-07	3.227e-04	2.867e-01

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO:025231564	-0.25231564	41	2.340e-08	6.285e-05	TMEM126A:20 TMEM1268:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223
Mitochondrial Respiratory Chain Complex	-0.25231564	41	2.340e-08	6.285e-05	TMEM126A:20 TMEM1268:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223
Mitochondrial Respiratory Chain Complex	-0.19480931	62	1.179e-07	2.110e-04	TMEM126A:20 TMEM1268:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223
Mitochondrial Gene Expression (GO:0140005	-0.15155407	91	6.201e-07	8.326e-04	PTCD3:27 MRPL37:33 MTIF2:175 MRPL43:176 RARS:225 MRPS30:226 FASTKD5:290
Mitochondrial Translation (GO:0032543)	-0.15186334	87	1.034e-06	1.111e-03	PTCD3:27 MRPL37:33 MTIF2:175 MRPL43:176 RARS:225 MRPS30:226
tRNA Methylation (GO:0030488)	-0.21803620	34	1.010e-05	9.856e-03	TRMT9B:145 TRMT10C:184 METTL4:444 THUMPD2:483 TWY5:517 TRMT10B:598
Oxidative Phosphorylation (GO:0006119)	-0.18085227	41	6.255e-05	4.799e-02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA3:353 ATP5B:368 TEFM:423
Mitochondrial Electron Transport, NADH T	-0.22504045	25	9.591e-05	6.439e-02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 NDUFB4:424 NDUFB4:464
Aerobic Respiration (GO:0009060)	-0.17835275	39	1.178e-04	7.033e-02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 TEFM:423 NDUFB4:424
Cell Junction Assembly (GO:0034329)	-0.13896227	63	1.398e-04	7.510e-02	SHANK3:7 SHANK2:62 SDK1:143 ARVCF:147 DBNL:169 SDK2:368
Calcium Ion Transport (GO:0006816)	-0.11348289	93	1.613e-04	7.875e-02	ITPR1:24 TRPM5:49 TRPM1:107 CACNA1S:201 CACNG3:266 CRACR2B:285
Proton Motive Force–Driven Mitochondrial	-0.17965487	36	1.937e-04	8.670e-02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 ATP5B:368 NDUFB4:424
tRNA Modification (GO:0006040)	-0.13657067	61	2.301e-04	9.508e-02	ALKBH8:116 TRMT9B:145 METTL4:444 THUMPD2:483 DTWD1:504 TWY5:517
Proton Motive Force–Driven ATP Synthesis	-0.17174606	38	2.520e-04	9.666e-02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 ATP5B:368 NDUFB4:424
Cilium Assembly (GO:0060271)	-0.07339778	197	4.111e-04	1.472e-01	ARL13B:21 CEP126:24 FAM161A:31 TCTN3:40 GORAB:66 DNMBP:78
Calcium Ion Import Across Plasma Membran	0.17509270	73	5.046e-04	1.694e-01	CACNA1B:47 SCN2A:81 TRPM1:107 CACNA1S:201 TRPV6:300 CLU:396
mRNA 3'-End Processing (GO:0031124)	0.18425382	28	7.449e-04	2.278e-01	SLBP:3 CSTF2:14 CDT3:73 WDR33:304 SYMPK:386 PEK2:508
Muscle Contraction (GO:0006936)	0.11492762	72	7.635e-04	2.278e-01	CACNA1B:201 LMOD1:245 MYH2:322 EMD:399 MYH1:819 RYR2:833
Aerobic Electron Transport Chain (GO:001	-0.14145558	43	1.354e-03	2.535e-01	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 NDUFB4:424 NDUFB4:464
Cell–Cell Junction Assembly (GO:0007043)	0.12854013	53	1.226e-03	2.535e-01	ARVCF:147 JAM3:275 TJPI:478 TLN1:514 GJA4:584 TBCD:751
Cellular Respiration (GO:0045333)	-0.12947663	53	1.128e-03	2.535e-01	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 NDUFB4:424 NDUFB4:464
Cilium Organization (GO:0044782)	-0.06895408	183	1.369e-03	2.535e-01	ARL13B:21 CEP126:24 FAM161A:31 TCTN3:40 DNMBP:78 CDDC66:105
Glycogen Metabolic Process (GO:0005977)	0.23932592	15	1.335e-03	2.535e-01	PYG8:101 PER2:268 PYGM:597 GYS1:1044 GYS2:1104 GYS2:1280
Negative Regulation Of Fibroblast Prolif	-0.17957067	43	1.345e-03	2.535e-01	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 NDUFB4:424 NDUFB4:464
Mitochondrial RNA Processing (GO:0000963	-0.31631748	9	1.018e-03	2.535e-01	SUPV3L1:46 TRMT10C:184 FASTKD5:290 PNPT1:511 FASTKD2:1042 FASTKD1:1105
Negative Regulation Of Fibroblast Prolif	0.28705643	11	9.806e-04	2.535e-01	CDC73:73 MYC:338 EMD:399 PEK2:508 TRIM32:1520 SKI:1905
Negative Regulation Of Transcription By	0.04598413	440	1.090e-03	2.535e-01	ZNF281:6 EHM1:33 ZBTB45:39 TCF25:54 TCF15:59 CDC73:73
Positive Regulation Of Cytosolic Calcium	0.20811836	20	2.179e-03	2.535e-01	OPRL1:112 FZRL1:217 GPR55:308 GRM1:418 GPR35:465 GPR20:510
tRNA Wobble Base Modification (GO:000209	-0.27510614	12	9.699e-04	2.535e-01	ALKBH8:116 TRMT9B:145 ELP2:182 ELP5:815 ELP4:886 MTOB:1101
DNA Repair (GO:0006281)	-0.05999278	234	1.683e-03	3.103e-01	BRCA1:16 RNF169:49 FANCM:81 RNF168:115 DCLRE1A:125 RNF21:144
Chemical Synaptic Transmission (GO:00072	0.06556424	188	2.040e-03	3.402e-01	CACNA1B:47 RIMBP2:76 GABRB3:78 KCNQ3:172 TH:254 RPS6KA2:299
mRNA Export From Nucleus (GO:000406)	0.14680814	37	2.018e-03	3.402e-01	MCM3AP:13 IWS1:230 NUP160:372 NUP133:442 ALYREF:592 THPOC5:634
Regulation Of Cilium Movement (GO:000335	-0.33490171	7	2.154e-03	3.402e-01	DNAH11:203 CFAF206:361 CFAF43:70 CYB5D1:725 CDDC65:820 CDDC40:6080
Synaptic Vesicle Uncoating (GO:0016191)	0.44346531	4	2.129e-03	3.402e-01	GAK:86 SH3GL2:498 SYNJ1:1067 SH3GL3:1101 NANA:1A
Regulation Of Long-Term Synaptic Potenti	0.20185110	19	2.328e-03	3.573e-01	SHANK3:7 ADCY1:25 NSG1:314 SHISA7:1002 YTHDF1:1346 ADCY8:1347
poly(A) ⁺ mRNA Export From Nucleus (GO:000	0.27720566	10	2.406e-03	3.589e-01	MCM3AP:13 IWS1:230 NUP133:442 ZC3H1A:1088 GEL1:1805 DDY25:2097
RNA Export From Nucleus (GO:0006405)	0.14017085	38	2.814e-03	3.637e-01	MCM3AP:13 NUP160:372 NUP133:442 ALYREF:592 NUP155:633 THOCS:634
Adenylate Cyclase–Inhibiting G Protein-C	0.14121403	37	2.979e-03	3.637e-01	OPRL1:112 GNAI3:239 HRH4:356 OPRK1:625 PDE2A:717 DRD2:761
Adherens Junction Organization (GO:00343	0.16239026	28	2.956e-03	3.637e-01	JAM3:275 INAVA:369 TJPI:478 MTS1:583 TBCD:751 CDH6:781
Cell–Cell Junction Organization (GO:0045	0.12348637	49	2.819e-03	3.637e-01	TGFB1:89 ARVCF:147 FZRL1:217 TJPI:478 TLN1:514 GJA4:584

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_TP22_AMPLICON	0.30081687	32	3.934e-09	2.548e-05	INTS1:71 ADAP1:154 EIF3B:181 TTYH3:249 GRIFIN:301 MAFK:353
WP_MITOCHONDRIAL_COMPLEX_1_ASSEMBLY_MODE	-0.25124301	38	8.478e-08	2.745e-04	TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223 NDUFB4:424
BENPORATH_ES_WITH_H3K27ME3	0.05798649	733	1.329e-07	2.870e-04	RG59:26 FRMPD4:42 CACNA1B:47 MMP2:48 SMPD3:58 COL12A1:75
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	0.39551650	14	3.001e-07	4.169e-04	EP40A:1 P2RX2:178 POLE:124 GOLGA3:378 ANKLE2:425 ULK1:578
REACTOME_COMPLEX_1_BIOGENESIS	-0.23659871	39	3.219e-07	4.169e-04	TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223 NDUFA9:353
REACTOME_DEVELOPMENTAL_BIOLOGY	0.05709612	679	5.420e-07	5.850e-04	MMP2:48 ARHGAP39:52 PTK2:63 TRIO:69 ABLIM2:72 SCN2A:81
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	-0.15805506	79	1.230e-06	1.138e-03	LRPPRC:42 TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223
PEREZ_TP53_TARGETS	0.05013360	813	1.694e-06	1.371e-03	TGFBRAPI:4 PHERF1:8 SEC14L5:15 MICAL3:18 BRD1:19 ADCY1:25
NIKOLSKY_BREAST_CANCER_SP15_AMPLICON	0.30743973	20	1.947e-06	1.401e-03	EXOC6:61 CEP72:191 LRRC14B:566 IRX4:621 BRD9:741 SLC6A18:884
REACTOME_INFECTION_DISEASE	0.05854877	555	3.013e-06	1.774e-03	ZDHHC8:5 ENTPD1:22 ITPR1:24 ADCY1:25 PTK2:63 NUP37:68
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT_	-0.16680093	66	2.848e-06	1.774e-03	LRPPRC:42 TMEM126B:55 NDUFS3:77 NDUFB8:95 NDUFB5:158 ECSIT:223
JOHNSTONE_PARVB_TARGETS_3_DN	-0.05761698	568	3.374e-06	1.821e-03	RTKN2:59 PKFB3:111 ARHGAP11A:114 ZWINT:118 SPAG5:122 LRR1:142
REACTOME_NEURONAL_SYSTEM	0.08110099	270	4.978e-06	2.480e-03	ADCY1:25 CACNA1B:47 SHANK2:62 GABRB3:78 DBNL:169 KCNQ3:172
GINESTIER_BREAST_CANCER_ZNF217_AMPLIFIED	0.07995508	253	1.299e-05	6.009e-03	TRAF2:2 BRD1:19 WDR33:304 SYMPK:386 SLU7:42 ALYREF:592
WP_Q237_COPY_NUMBER_VARIATION_SYNDROME	0.13416109	86	1.749e-05	7.550e-03	KIF1A:31 KHLH:30:57 TRAF3P1:106 DGKD:115 GPC1:116 MROH2A:126
MOOHTA_HUMAN_MITOD8_6_2002	-0.06851299	328	2.225e-05	9.005e-03	CPT2:9 SUPV3L1:46 ACADM:57 NDUFS3:77 NDUFB8:95 MLH3:147
REACTOME_MITOCHONDRIAL_TRANSLATION	-0.13495764	82	2.451e-05	9.336e-03	PTCD3:27 MRPL37:33 MTIF2:175 MRPL43:176 MRPS30:226 ERALL1:378
GEORGES_TARGETS_OF_MIR192_AND_MIR215	-0.04845097	655	2.908e-05	1.046e-02	WDCP:8 BRCA1:16 ATAD5:36 NIN:38 DST:54 RTNK:59
WP_OXIDATIVE_PHOSPHORYLATION	-0.02029660	35	3.274e-05	1.116e-02	NDUFS3:77 NDUFB8:95 NDUFB5:158 NDUFA9:353 ATP5B:368 NDUFB4:424
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	0.05814282	435	3.684e-05	1.142e-02	CADM3:32 ADGBR1:53 QTJ07A:104 MAPKBP2:111 KCNQ3:172 CNTN4:175
REACTOME_TRANSPORT_OF_MATURE_TRANSCRIPT_	0.15809903	57	3.705e-05	1.142e-02	SLBP:3 NUP37:68 NUP210:164 WDR33:304 NUP43:351 NUP160:372
NIKOLSKY_BREAST_CANCER_16P13_AMPLICON	0.13206946	78	5.633e-05	1.658e-02	CRAMP1:80 UNKL1:32 JPT2:267 CAPN15:328 RPU5D1:391 MEIOB:419
MARTENS_TRETINOIN_RESPONSE_UP	0.05009692	558	6.194e-05	1.744e-02	KIF1A:31 GJB4:34 SHANK2:62 SPON2:84 WNK2:97 CHST15:125
MIKKELSEN_NPC_ICP_WITH_H3K4ME3	-0.06590007	311	7.039e-05	1.850e-02	WDCP:8 KHLH:5:23 A2M:25 ALKBH8:116 DCLRE1A:125 RBMS2:135
REACTOME_PROCESSING_OF_CAPPED_INTRON_CON	0.08880110	169	7.141e-05	1.850e-02	SLBP:3 CSTF2:14 NUP37:68 NUP210:164 SUGP1:282 WDR33:304
REACTOME_DISEASES_OF_DNA_REPAIR	-0.16662397	47	7.833e-05	1.951e-02	BRCA1:16 RAD51AP1:170 ATM:244 MUTHY:362 RMI1:429 WRN:438
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.07455828	230	1.043e-04	2.501e-02	CADM3:32 ADGBR1:53 STUM:60 PTGFRN:88 TGFBI:89 GJA3:144
MIKKELSEN_NPC_HCP_WITH_H3K27ME3	0.07268223	241	1.087e-04	2.513e-02	CADM3:32 ADGBR1:53 TCO4C1:182 CTR2:187 TRAP2D:236 KCNQ2:343
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	0.05711356	389	1.228e-04	2.743e-02	MMP2:48 ARHGAP39:52 PTK2:63 TRIO:69 ABLIM2:72 SCN2A:81
REACTOME_SIGNALING_BY_RECEPTOR_TYROSINE_	0.03284286	376	1.285e-04	2.774e-02	ITPR1:24 PTK2:63 GABRB3:78 EPN1:140 PTEN:141 ADAP1:154
NIKOLSKY_BREAST_CANCER_Q2012_Q13_AMPLICO	0.11045687	100	1.385e-04	2.894e-02	TCFL5:59 PFKFB3:111 CABLES2:137 ZBTB46:186 KCNG12:152 SYCP2:229
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.17722494	37	1.925e-04	3.896e-02	SLBP:3 CSTF2:14 WDR33:304 SYMPK:386 SLU7:42 ALYREF:592
REACTOME_TRANSPORT_OF_MATURE_MRNAS_DERIV	0.19191232	31	2.184e-04	4.286e-02	SLBP:3 NUP37:68 NUP210:164 WDR33:304 NUP43:351 NUP160:372
BENPORATH_EED_TARGETS	0.04194902	676	2.389e-04	4.421e-02	PARP12:3 TRAF2:2 CACNA1B:47 COL12A1:75 KHLH:30:57 ZCCHC2:108
EPPERT_PROGENITOR	-0.10402122	105	2.364e-04	4.421e-02	LRPPRC:42 SORD:97 ZWINT:118 HAUS6:289 DCL1:350 NET1:391
REACTOME_DNA_DOUBLE_STRAND_BREAK_REPAIR	-0.09957657	113	2.623e-04	4.719e-02	BRCA1:16 RNF168:115 RAD51AP1:170 POLQ:213 MUS81:243 ATM:244
MOOHTA_MITOCHONDRIA	-0.05471765	337	3.125e-04	5.470e-02	SUPV3L1:46 ACADM:57 METTL1:761 NDUFS3:77 NDUFB8:95 NDUFB5:158
WP_DNA_REPAIR_PATHWAYS_FULL_NETWORK	-0.04749778	98	3.443e-04	5.717e-02	BRCA1:16 FANCM:81 FANCD1:331 ATM:244 MUTHY:362 XRCC4:364
WONG_MITOCHONDRIA_GENE_MODULE	-0.08469153	151	3.397e-04	5.717e-02	LRPPRC:42 NDUFS3:77 NDUFB8:95 NDUFB5:158 MRPS30:226 NDUFA9:353
KEGG_CALCIIUM_SIGNALING_PATHWAY	0.08222773	152	4.828e-04	7.817e-02	ITPR1:24 ADCY1:25 CACNA1B:47 P2RX2:178 ADCY9:179 PTK2B:193

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Mitochondrial Diseases	-0.07709801	308	3.918e-06	3.827e-02	CPT2:9 SARS2:19 TMEM126A:20 LRPPRC:42 TMEM126B:55 METTL1:761
Increased CSF lactate	-0.17700443	44	4.944e-05	2.414e-01	LRPPRC:42 TMEM126B:55 NDUFS3:77 TRMT10C:184 RARS:225 NDUFA9:353
Autism Spectrum Disorders	0.05744652	369	1.749e-04	4.271e-01	SHANK3:7 HERC2:21 ITPR1:24 EHM1:33 SHANK2:62 JAKMIP1:67
Schizophrenia	0.03284286	1277	1.431e-04	4.271e-01	ZDHHC8:5 SHANK3:7 BRD1:19 HERC2:21 ADCY1:25 RGS9:26
Cataract, Pulverulent	0.43837711	5	6.871e-04	5.965e-01	GJA3:144 BFSP2:205 GJA8:63s CRYBB1:900 HSF4:1858 NA
Acute neurodeficiency encephalopathy	-0.25728441	16	3.679e-04	5.965e-01	TMEM126B:55 NDUFS3:77 FOXRED1:480 TIMMDC1:591 NDUFA11:642 NDUFAF4:695
Gene recurvatum	0.27460241	13	6.093e-04	5.965e-01	CTDP1:74 COL1A2:462 CANT1:820 FBN1:882 COL5A1:1241 XYLT1:1391
Horizontal eyebrow	0.36758899	7	7.580e-04	5.965e-01	DEAF1:849 RERE:956 EBF3:1474 PROM1:651 SKI:1905 TRAPPC9:2338
Hyperchogenic kidneys	-0.43451588	5	7.661e-04	5.965e-01	SARS2:19 CEP290:328 TCTN2:503 INVS:1540 REN:1864 NA
Malignant neoplasm of fallopian tube	-0.35300461	8	5.458e-04	5.965e-01	BRCA1:16 CASCI1:241 ATM:244 NBR1:499 BARD1:779 BRCA2:1199
NADH:Q(1) Oxidoreductase deficiency	-0.21186539	22	5.849e-04	5.965e-01	TMEM126B:55 NDUFS3:77 FOXRED1:480 TIMMDC1:591 NDUFA11:642 NDUFAF4:695
Pallor of optic disc	-0.14312575	46	7.938e-04	5.965e-01	TMEM126A:20 ARL13B:21 FAM161A:31 TMEM126B:55 NDUFS3:77 CERKL:323
Progressive macrocephaly	-0.23926967	19	3.070e-04	5.965e-01	TMEM126B:55 NDUFS3:77 FOXRED1:480 TIMMDC1:591 NDUFA11:642 NDUFAF4:695
Aneurysm, Ruptured	0.36124569	7	9.344e-04	6.519e-01	RNH1:231 PPP1R13L:64 NOS3:646 SHC3:959 FIN1:1239 ACE:1710
Central neuroblastoma	0.02983452	1085	1.290e-03	7.414e-01	HERC2:21 ITPR1:24 KIF1A:31 ZNF236:41 MMP2:48 PTK2:63
Huntington Disease	0.04723750	403	1.285e-03	7.414e-01	FAM193A:20 ITPR1:24 APC2:36 MMP2:48 TGFBI:89 HSF1:138
Slender finger	0.29552897	10	1.214e-03	7.414e-01	COL12A1:75 COL6A1:165 COL6A2:255 ZNF592:969 LFNG:1177 KANSL1:1448
Non-obstructive azospermia	0.12889483	51	1.470e-03	7.977e-01	ENTPD6:29 BOLL:206 WDR11:295 DAZL:359 KITLG:359 MEIOB:419
CATARACT, COPPOCK-LEIKE	0.36324971	5	4.910e-03	8.476e-01	GJA3:144 BFSP2:205 GJA8:63s CRYBB1:900 ISO2:6651 NA
COLORBLINDNESS, PARTIAL, DEUTAN SERIES	0.17549051	24	2.948e-03	8.476e-01	TGFB1:89 ANXA6:237 LMOD1:245 PLYNA4:630 CCR5:667 NPM1M:1091
Epilepsy, Temporal Lobe	0.07286175	122	5.910e-03	8.476e-01	GABRB3:78 SCN2A:81 LETM1:150 KCNQ3:172 BIN1:207 SOD1:302
Hyperkinesia, Generalized	0.18819478	21	3.151e-03	8.476e-01	ADCY1:25 TH:254 CNTN